

Optimization of the European interconnected electrical network using a parallelized heuristic algorithm

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Abstract. Generating electricity from fossil fuels has serious environmental consequences: pollution, overexploitation of resources, and accelerated climate change. This is why a global transition to renewable energy is essential. However, their intermittent and unpredictable nature often raises doubts about their reliability, especially in the face of increasing demand in future scenarios. Currently, systems of electricity monitoring have been modelled by linear optimization problems in order to find a solution to the intermittency of these sources. Some are written for the French electrical network, and another one exists for the European network, including the exchange of electricity between countries. The complexity of the linear programming algorithms needed to solve these systems is considerable, especially for a European optimization, since it increases more than proportionally with the number of variables. This study aims to develop an heuristic approach and a computationally efficient algorithm to approximate the solution at the European level. It includes imports and exports, and an individual optimization of storage in each country. The methodology is based on constrained linear programming. A two-step development process was adopted: the algorithm was first refined on a simplified model, allowing for rapid iterations, before being extended to a comprehensive framework. This served as a model for comparing the results of the tested heuristic algorithms. We focused on one of them, which we detail in this report. At the end of our study, our results are quite conclusive. We did manage to reduce execution time, and the result obtained is not too far from the reference. We conclude that we took an interesting direction and that we should keep observing the behaviour of this algorithm with different input : realistic data, new technologies or extreme meteorologic events.

Introduction. The use of fossil fuels has serious consequences for the environment and climate. It is becoming urgent to limit their use and rethink our production methods towards sustainable solutions. Yet, even today, many are wary of "green" energies because they are intermittent. It is therefore crucial to consider how to optimize electricity storage facilities and exchanges between European countries to successfully eliminate the use of fossil fuels in times of shortage and ultimately transition to a 100% renewable energy mix.

In recent years, several researchers have focused on electricity grid management. For example, Philippe Quirion and his team developed the EOLES models [4][3] to explore electricity strategies in France and Europe up to 2050. This work is based on RTE data concerning production and storage, as well as ADEME consumption scenarios. Furthermore, the serious game Climix, designed by our supervisors, aims to raise public awareness of the challenges of the energy transition.

Despite these advances, it remains a barrier in program execution time, especially when working on a model with interconnected countries. However, it is the model that brings the most "flexibility" solutions compared to a one-country model because it adds imports and exports to storage solutions. We can show that the number of variables for N countries is $7 \times N + (N - 1)$ for a toy model. The complexity is therefore quadratic. This limits the possibility of testing a large number of scenarios. This is why we need to find less time-

consuming ways to explore various scenarios and solutions.

The objective of this study was therefore to design a fast and efficient algorithm capable of integrating electricity imports and exports on a European scale. We first formulate the optimization problem using Linear Programming on a reduced version of the full problem, which serves as a theoretical benchmark. This reduced problem is then decomposed into two sub-models - one for national energy storage and one for cross-border flows - combined through task parallelization. After that, we compare the heuristic solution against the theoretical benchmark in terms of run-time and solution quality.

This paper is organized as follows. [section 1](#) details the complete problem. Firstly, in its complete form extracted from the multi-country EOLES problem, and then in its reduced form. It formulates the optimization problem using Linear Programming, which serves as a theoretical benchmark. [section 2](#) is about the heuristic approach. It describes the mechanism and the two sub-models. Then, [section 3](#) shows the results of the simulations and comparison between the two approach. It compares the heuristic solution against the theoretical benchmark in terms of run-time and solution quality.

1. Description of the complete problem.

1.1. The interconnected EOLES model. The EOLES model family (Energy Optimization for Low Emission Systems), developed by P. Quirion and collaborators [4], is a modelling tool for the French electricity system. We work on a particular model that only considers renewable production technologies (offshore and onshore wind, photovoltaic (PV) solar energy, lake energy, run-of-river energy, and biogas) and three electricity storage technologies (pumped-storage hydropower plants (PHS), batteries, and methanation (PTG)) (cf. [Figure 1](#)). Its objective is to identify an electricity mix capable of satisfying hourly electricity demand in France in 2050, while minimizing production and storage costs subject to constraints. The model relies on linear optimization in which France is represented as a single node, with no regional distinction, and where only metropolitan territories are considered. Electricity exchanges with foreign countries are not taken into account, which requires demand to be met at every hour since the model does not allow electricity deficits (or shortages). Furthermore, storage capacities at the end of the year must be at least as full as at the beginning of the year. Finally, demand for each hour of the year is determined before launching the optimization; it is a known input of the problem.

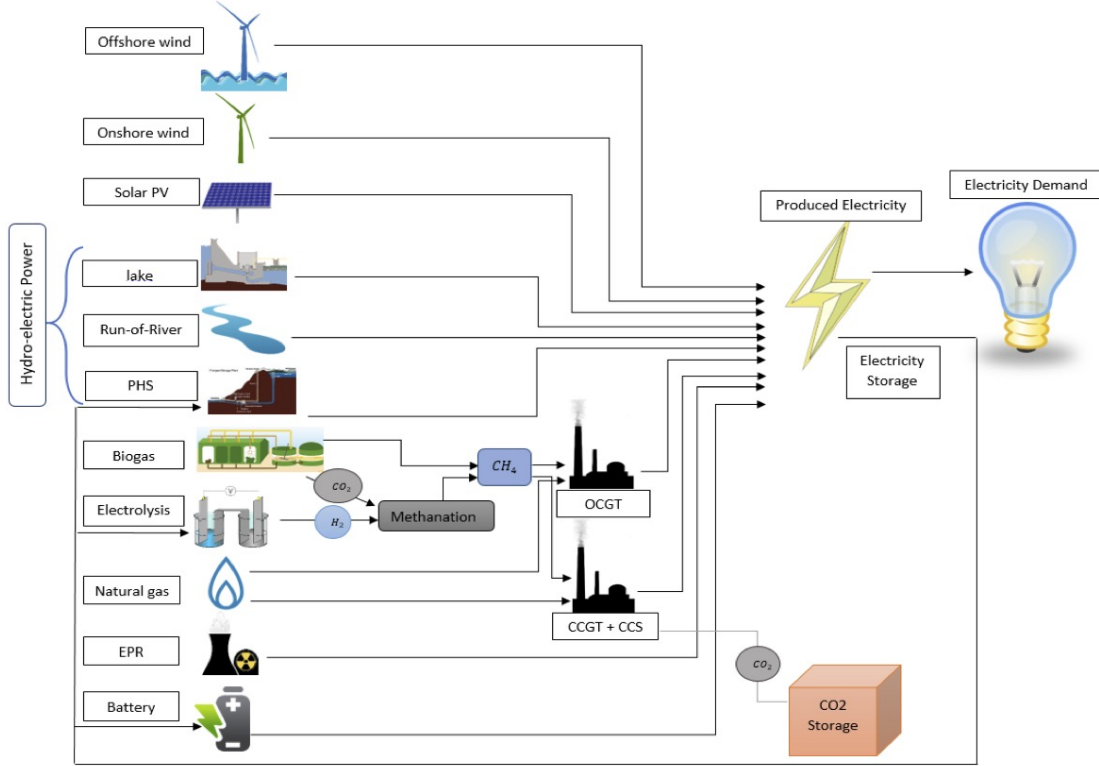


Figure 1: Electricity production sources and storage devices [4].

The model of interest to us is the N -country model. It will allow us to carry out optimizations at the European scale. For a given objective (for example, the minimization of shortages), we will seek to approach it optimally for each country considered, including the possibility for each country to import or export surplus electricity.

First, here are the model parameters we will use to calculate the number of variables to optimize:

- $N = 7$: number of countries considered in the model
- $N_{thr} = 9$: number of so-called "thermal" electricity production technologies (nuclear, gas, coal...)
- $N_{str} = 3$: number of electricity storage technologies (PHS, batteries, methanation)
- $N_{tec} = 17$: number of electricity production and storage technologies (including thermal)
- $N_{exo} = 11$: number of countries outside the model

We now present the variables used in this model. The following table presents the hourly variables defined for a country c in the N -country EOLES model. Furthermore, throughout the table: $tec \in \llbracket 1, N_{tec} \rrbracket$, $thr \in \llbracket 1, N_{thr} \rrbracket$, $str \in \llbracket 1, N_{str} \rrbracket$, $c' \in \llbracket 1, N - 1 \rrbracket$, $c_{exo} \in \llbracket 1, N_{exo} \rrbracket$.

Table 1: Hourly variables for country c in the N -country EOLES model

Variables to optimize	Description
$gene_{tec}$	Hourly electricity production in GWh/h
on_{thr}	Available electrical capacity in GW
$start_up_{thr}$	Capacity started at hour t in GW
$turn_off_{thr}$	Capacity shut down at hour t in GW
$ramp_up_{thr}$	Positive increase in electricity produced in GW
$storage_{str}$	Incoming storage in GW
$stored_{str}$	State of charge in GWh
rsv_{tec}	Capacity in reserve for grid stabilisation in GW
hll	Unserved load (shortage) in GW
$im_{c'}$	Imports from country c' in the model in GW
$ex_{c'}$	Exports to country c' in the model in GW
$exo_im_{p_{exo}}$	Imports from countries exogenous to the model in GW
$exo_ex_{p_{exo}}$	Exports to countries exogenous to the model in GW
$hcost$	Cost of exchanges between countries in thousands of euros
$hcarb$	CO2 emissions in tCO2

Let N_c be the number of variables to optimize at hour t for country c , we have:

$$\begin{aligned}
(1.1) \quad N_c &= \{(gene_{tec})_{tec}\} + \{(rsv_{tec})_{tec}\} \\
&+ \{(on_{thr})_{thr}\} + \{(start_up_{thr})_{thr}\} + \{(turn_off_{thr})_{thr}\} + \{(ramp_up_{thr})_{thr}\} \\
&+ \{(storage_{str})_{str}\} + \{(stored_{str})_{str}\} \\
&+ \{(im_{c'})_{c'}\} + \{(ex_{c'})_{c'}\} \\
&+ \{(exo_im_{p_{exo}})_{p_{exo}}\} + \{(exo_ex_{p_{exo}})_{p_{exo}}\} \\
&+ \{hll\} + \{hcost\} + \{hcarb\} \\
&= 2 \times N_{tec} + 4 \times N_{thr} + 2 \times N_{str} + 2 \times (N - 1) + 2 \times N_{exo} + 3 \times 1 \\
&= 2 \times 17 + 4 \times 9 + 2 \times 3 + 2 \times 6 + 2 \times 11 + 3 \times 1 \\
&= 113
\end{aligned}$$

Therefore, the total number of variables to optimize for this problem with $N = 7$ countries over a one-year period is:

$$(1.2) \quad N_{total} = N_c \times N \times 8760 = 113 \times 7 \times 8,760 = 6,929,160$$

Simulating a solution to this problem over 1 year amounts to optimizing almost 7 million variables.

When the number of countries N increases, the number of variables grows sharply, leading to a considerable increase in computation time. Faced with this constraint, our objective is the following: find another way to formulate the problem in order to obtain results sufficiently close to those given by the N -country EOLES model, while reducing simulation time as much as possible.

To do so, we will work from a toy model that will be a lightweight version of the N -country EOLES model, enabling comparison of the results obtained. Ultimately, the goal is to apply our method, developed on the toy model, to the N -country EOLES model.

1.2. Data, variables and sources. For this study, we needed data concerning France and four countries that trade electricity with it, namely Germany, Italy, Spain, and the United Kingdom.

We used three kinds of dataset for each country in order to model the production of electricity and the demand for electricity :

- A demand dataset made of the electricity needed from a country at every hour.
- A solar production dataset composed of the solar power load factor for every hour.
- A wind production dataset consisting of the wind power load factor for every hour.

These datasets contain a piece of data for every hour of a year, which represents 8,760 variables for every dataset, 26,280 variables for every country, and 131,400 variables in total.

In addition to this data, we used an exchange capacity dataset that includes the maximum capacity of electricity that can be exchanged between two countries each hour. We modified it in order to create links between countries by expecting that in the future, those countries will exchange electricity. We added a link between Germany and the United Kingdoms, Germany and Italy and Italy and Spain.

We found those datasets in Philippe Quirion’s multi-country EOLES model code, freely available on GitHub [1].

Wind and solar power models the production of electricity for every country. It is the first kind of electricity sources that can answer the demand for a country. The other kind of sources is discharging the electricity that is stored in dams PHS for short-time storage and methanation PTG for long-term storage. The third kind is to import from a country that has surplus electricity. This implies that at each hour, after receiving the amount of electricity produced by the sun and the wind, the model decides whether it needs to discharge the PHS or PTG storage, or it imports electricity if there are shortages, or conversely, to recharge PHS or PTG, or export electricity to other countries in the case of surplus.

The following table 2 displays the fixed parameters and variables used in this model.

Table 2: Data and variables of the complete problem

Data (known parameters)	Description
N	Number of countries
T	Number of time steps
σ	Deficit penalty
$\text{capacity}_{str,c}$	Country's storage capacity per storage technology str
$\text{powerin}_{str,c}^{\max}, \text{powerout}_{str,c}^{\max}$	Country's maximum input/output power per storage technology str
$\text{efficiency}_{str,c}$	Country's conversion efficiency per storage technology str
η	Electricity exchange efficiency
$\text{fillrate}_{str,c}^0$	Initial fill rate per storage technology str ($\in [0, 1]$)
prod_c^t	Available production in the country c per hour t
demand_c^t	Energy demand in the country c per hour t
$q_{c,c'}^{\max}$	Maximum energy exchange capacity of country c to c'

Variables to optimize	Description
$\text{level}_{str,c}^t$	Storage level of country c at time t per storage technology str
$\text{powerin}_{str,c}^t, \text{powerout}_{str,c}^t$	Country's charge/discharge power per hour t per storage technology str
deficit_c^t	Electricity deficit in the country c per hour t
surplus_c^t	Surplus electricity in the country c per hour t (wasted electricity)
$q_{c,c'}^t$	Electricity exchanged from country c to c' at hour t

1.3. Creation of the theoretical benchmark. The first step was to create a simpler version of the problem that we can run in a reasonable amount of time. We obtain this complete problem :

$$\min \left(\sum_{c=1}^N -\text{level}_{\text{PTG},c}^{T-1} + \sigma \cdot \sum_{t=1}^T \text{deficit}_c^t \right),$$

under the constraints, $\forall c \in \{1, \dots, N\}, \forall str \in \{\text{PHS}, \text{PTG}\}$:

$$\begin{aligned} \text{level}_{str,c}^0 &= \text{capacity}_{str,c} \times \text{fillrate}_{str,c}^0, \\ \text{powerin}_{str,c}^t &\leq \text{powerin}_{str,c}^{\max}, \\ \text{powerout}_{str,c}^t &\leq \text{powerout}_{str,c}^{\max}, \\ \text{level}_{str,c}^t &\leq \text{capacity}_{str,c}, \\ q_{c,c'}[t] &\leq q_{c,c'}^{\max}, \\ \text{prod}_c^t - \text{demand}_c^t + \sum_{str} (\text{powerout}_{str,c}^t \times \text{efficiency}_{str} - \text{powerin}_{str,c}^t) \\ &\quad - \sum_{c'=1}^N q_{c,c'}^t + \sum_{c'=1}^N q_{c',c}^t \\ &= \text{surplus}_c^t - \text{deficit}_c^t, \\ \text{level}_{str,c}^t &= \text{level}_{str,c}^{t-1} \\ &\quad + \text{powerin}_{str,c}^t \times \text{efficiency}_{str,c} - \text{powerout}_{str,c}^t. \end{aligned}$$

This formulation by penalization enables putting the priority on minimizing the global deficit at each hour. When we approach the solution, it chooses one that maximizes the final gas stock. Then, we established constraints on :

- initial PTG and PHS stock rates ;
- power and capacity limits for storage (PTG and PHS) and exchanges ;
- energy balance : the value of surplus (if it is positive) or deficit (otherwise) must be equal to the remaining amount of electricity after its use, trade, and storage ;
- changes in stock level at each time step.

We used the library PuLP in Python to solve this linear problem and obtain a result in about one minute.

1.4. Simulations. Below you can see graphs obtained from the simulations. These results will serve as our reference for the rest of the process.

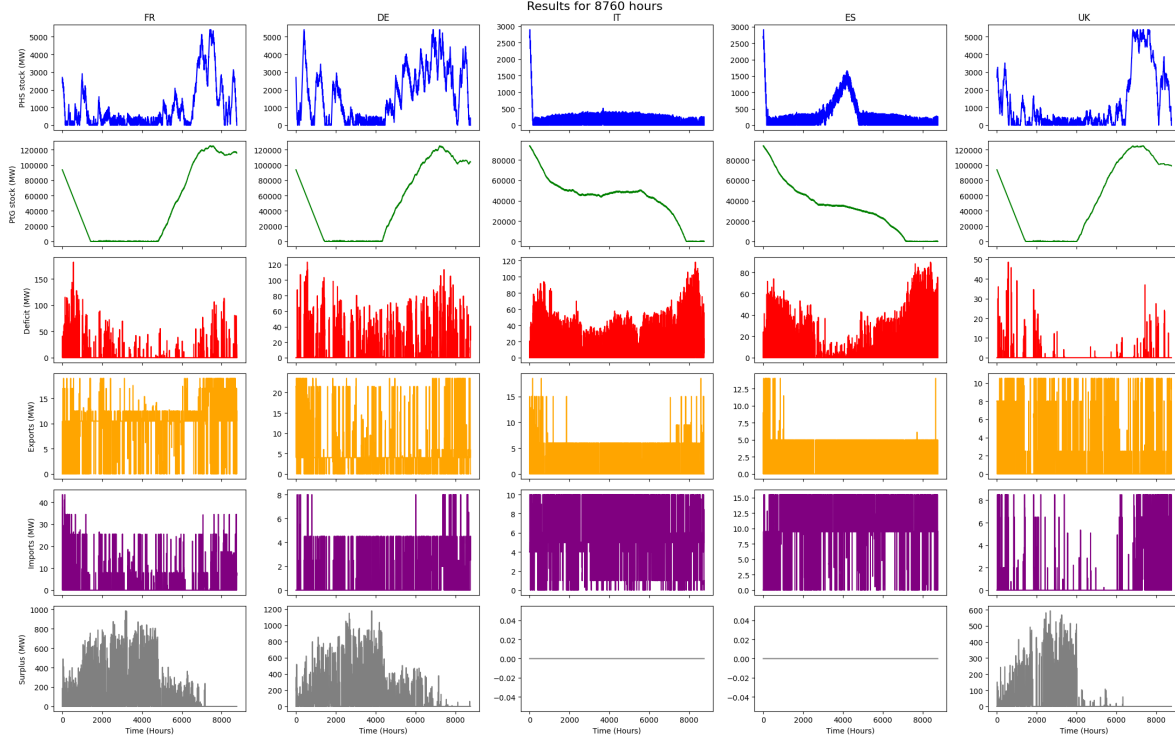


Figure 2: Optimization results - PHS level (blue), PTG level (green), deficit (red), exports (yellow), imports (purple) and surplus (gray) for each country over one year

It's important to remember that this data is simulated; in reality, the deficit is not as significant. Several things stand out in Figure 2. First, the seasonal patterns are quite clear: in winter (the beginning and end of the year), the level of PTG steadily declines, eventually depleting completely in some countries. This is because electricity demand is higher in winter, and solar power provides less electricity. Countries therefore use their stored electricity. In a real-world scenario, countries that completely exhaust their reserves should study their storage methods and increase their capacity. Second, the PHS stock fluctuates considerably, which is normal since it represents a source of short-term energy storage that can be charged and discharged rapidly. Then, we observe that the imports and exports graphs attain "levels" as if those value were discrete. Those levels are due to fact that the exports are capped by maximum exchange capacity. Finally, we observe that some countries have periods with large surpluses even though their stocks are not full. This is because optimization occurs all at once over the entire period, and since maximum storage is reached later, whether the country replenishes the stock at some point or another, the result is the same.

OPTIMIZATION OF THE EUROPEAN INTERCONNECTED ELECTRICAL NETWORK USING A PARALLELIZED HEURISTIC ALGORITHM

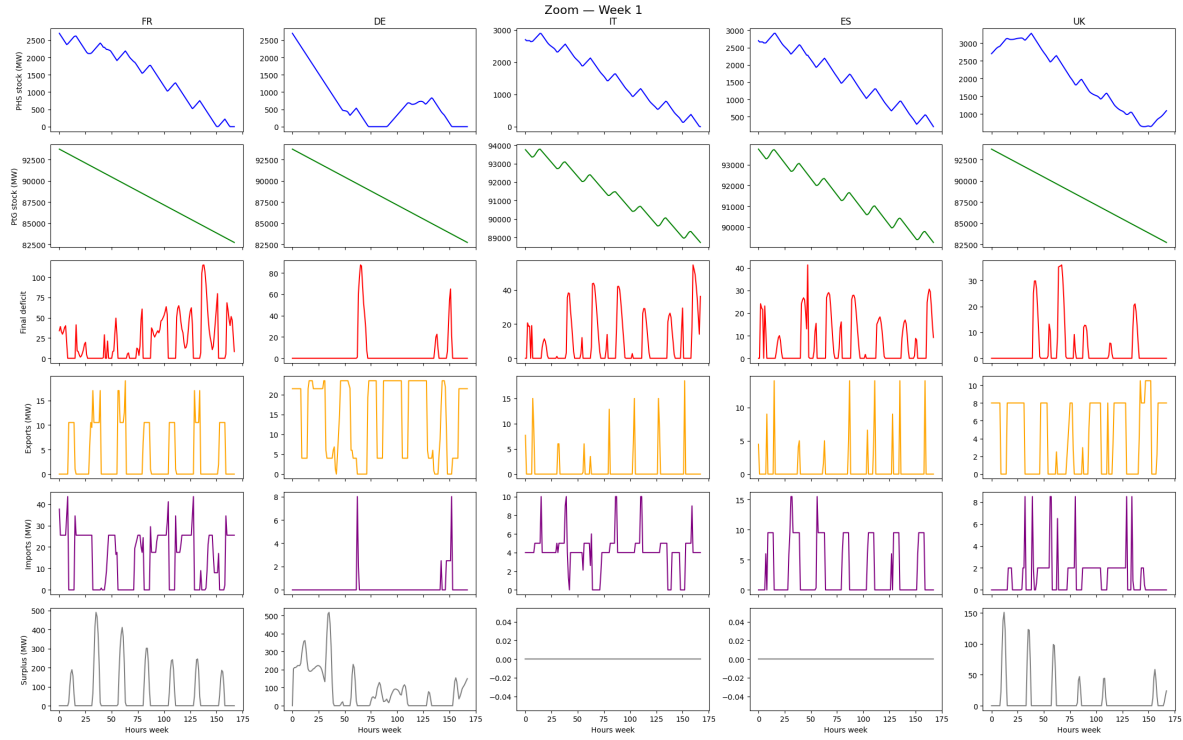


Figure 3: Optimization results - PHS level, PTG level, deficit, imports, exports and surplus for each country over one week

In Figure 3, we can see a zoomed-in view of the first week of the year. We notice that, for countries with a surplus, this surplus has peaks, representing the electricity supplied by solar power during the day. This means that the storage capacity is insufficient to capture all the electricity produced or that the storage technologies cannot convert electricity fast enough. Conversely, the deficits occur during the nighttime periods.

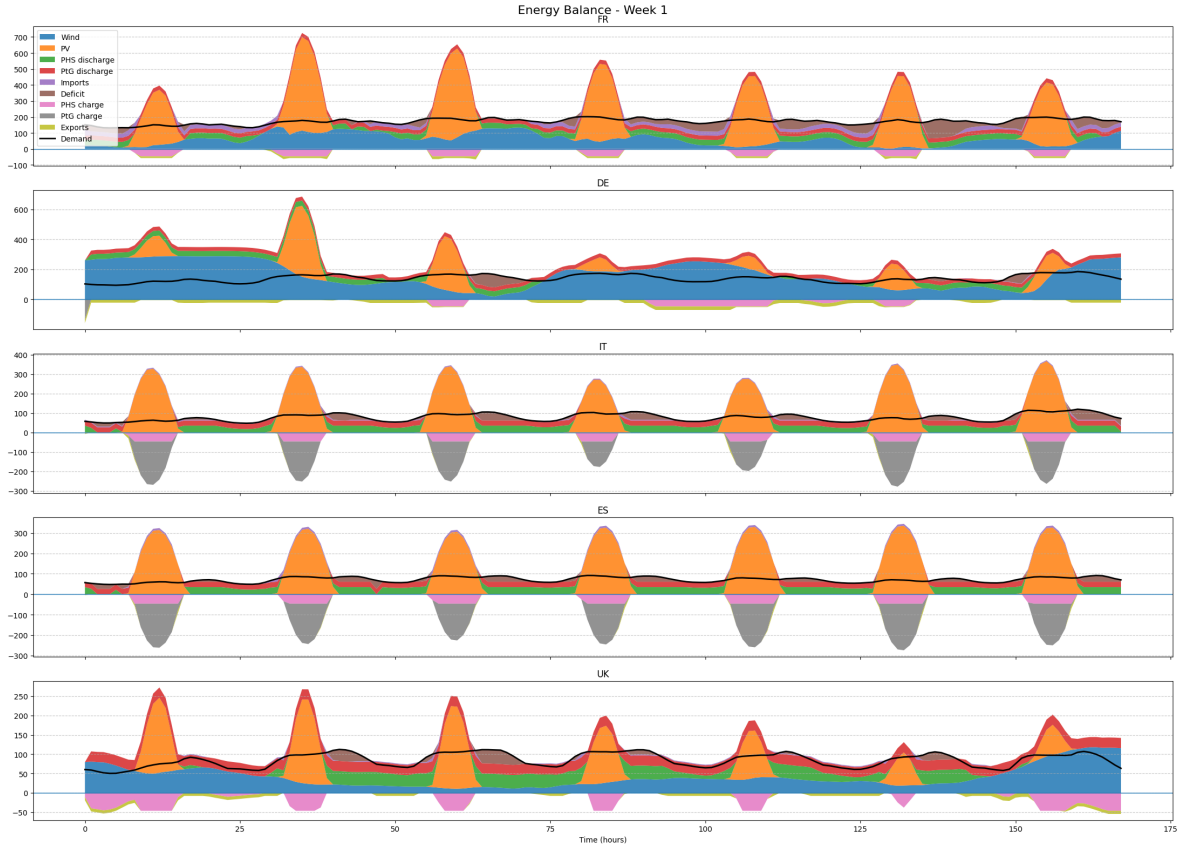


Figure 4: Cumulative graph of electricity quantities and demand curve - week 1

Finally, in Figure 4 we can see the same information but cumulatively over a week. This allows us to clearly see the electricity received and sent by all means in order to reach the demand curve. We can observe that some countries discharge their storage in order to provide electricity to countries in deficit.

2. The heuristic approach.

2.1. Description of our approach. The strategy used is as follows :

The first step is to calculate, for each country c , a tolerance interval $[-P_c^-, P_c^+]$ with $P_c^- = \sum_{str} \text{powerin}_{str,c}^{\max}$ and $P_c^+ = \sum_{str} \text{powerout}_{str,c}^{\max}$.

The second step is to optimize the exchanges between the countries 2.2. Each country initially possesses a quantity of electricity in $\mathbb{Z}$ equal to $prod_c - demand_c$. The objective is that, after the exchanges, as many countries as possible have this quantity within their tolerance interval. This means that, after the exchanges, each country possesses enough electricity to meet its demand after its storage is depleted, without having too much so that it can store any excess. This avoids shortages and prevents electricity waste. Subsequently, we will consider the electricity lost due to insufficient power or storage capacity as surplus.

The third step is to optimize internal storage for each country 2.3. This step can be

parallelized because each country performs its optimization internally. They no longer need to communicate with each other. This is where we hope to save program execution time.

2.2. Trade problem. This problem focuses on an hour t , and the goal is to optimize electricity exchange between countries.

Each country initially has a certain amount of available electricity (positive or negative) and a predefined tolerance margin. The maximum power levels that countries can exchange are also known. The problem is to determine the optimal import/export quantities so that as many countries as possible have their electricity available within their tolerance margin, while avoiding excessive and unnecessary exchanges. To make it easier to read, we have introduced the following notations, for all c in $\{1, \dots, N\}$:

$$\begin{aligned} \text{available}_c &:= \text{prod}_c - \text{demand}_c, \\ r_c &:= \text{available}_c - \sum_{c' \neq c} q_{c,c'} + \eta \sum_{c' \neq c} q_{c',c}. \end{aligned}$$

We also define the variable under_c which represents the deficit below P_c^- and the variable over_c which represents the surplus above P_c^+ :

$$\begin{aligned} \text{under}_c &:= \text{deficit}_c - P_c^-, \\ \text{over}_c &:= \text{surplus}_c - P_c^+. \end{aligned}$$

They allow for the creation of flexible constraints.

Here is the definition of the linear problem:

$$\begin{aligned} \min_{q, \text{over}, \text{under}} \quad & \sum_{c=1}^N \left(\sigma \cdot (\text{under}_c + \text{over}_c) + \sum_{c' \neq c} q_{c,c'} \right) \\ \text{under constraints, } \forall c \in \{1, \dots, N\} : \quad & \end{aligned}$$

$$\begin{aligned} r_c - \text{over}_c &\leq P_c^+, \\ r_c + \text{under}_c &\geq -P_c^-, \\ 0 &\leq q_{c,c'} \leq \text{max}_{c,c'} \\ \text{over}_c, \text{under}_c &\geq 0 \end{aligned}$$

For our problem, we have to do the optimization for each t in $\{1, \dots, T\}$. You can see in the following figures an example of results obtained in 5 countries, all connected, without storage technologies (i.e., $P_c^- = P_c^+ = 0$).

Deficits before and after trade by country

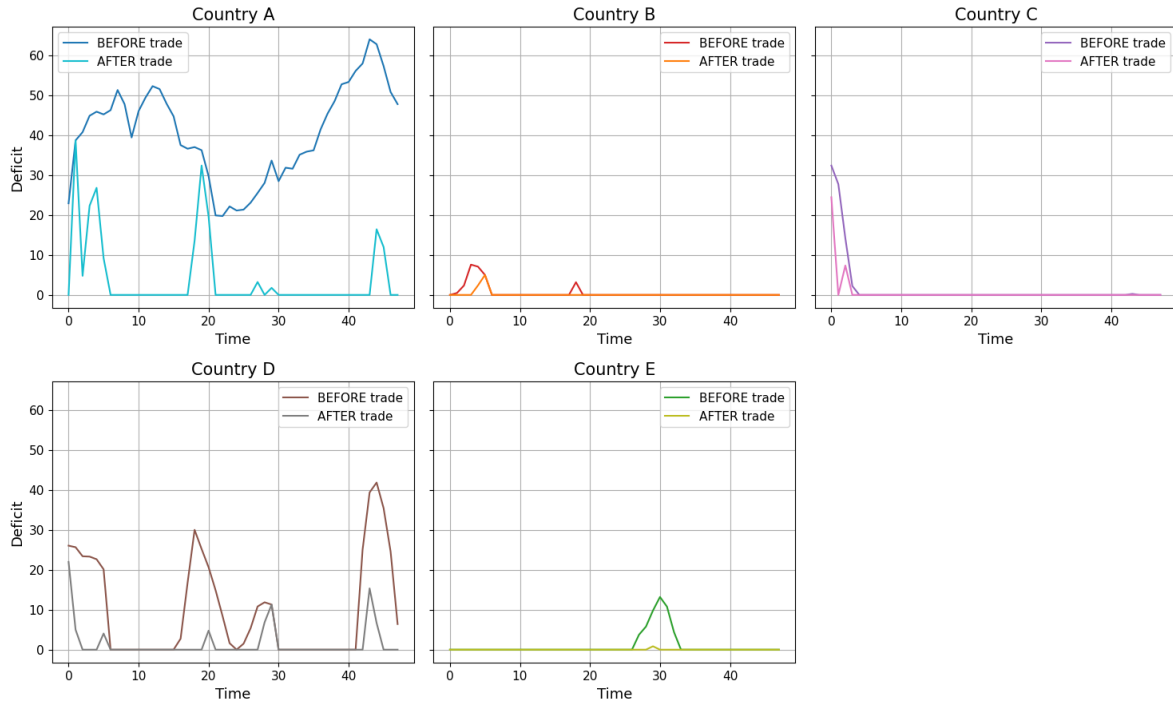
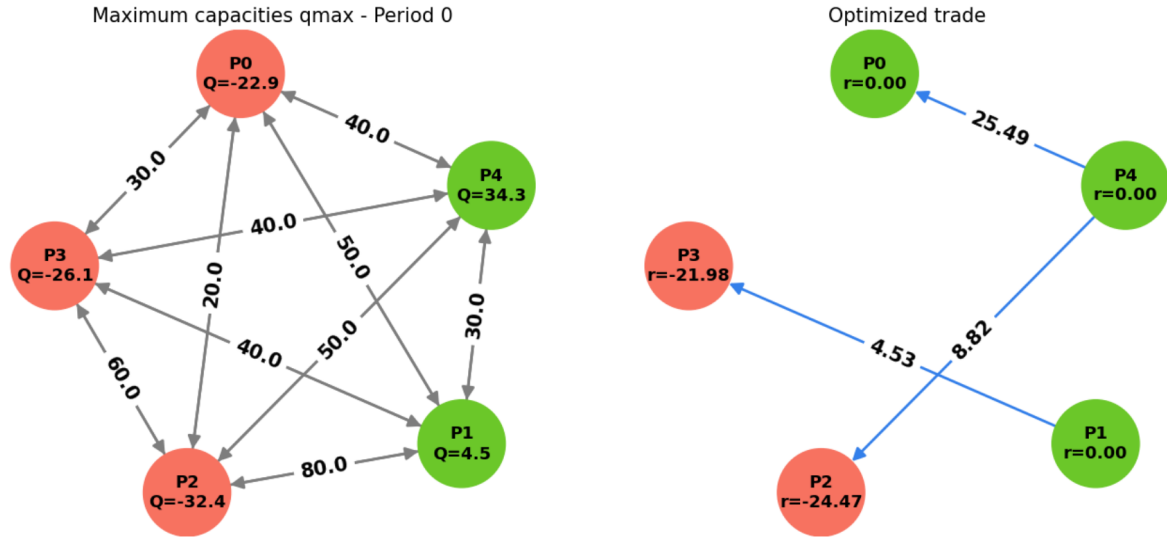


Figure 5: Examples from five countries: deficits before and after trade


Figure 6: Example 1 with 5 countries: Trade graphs at one hour t_1

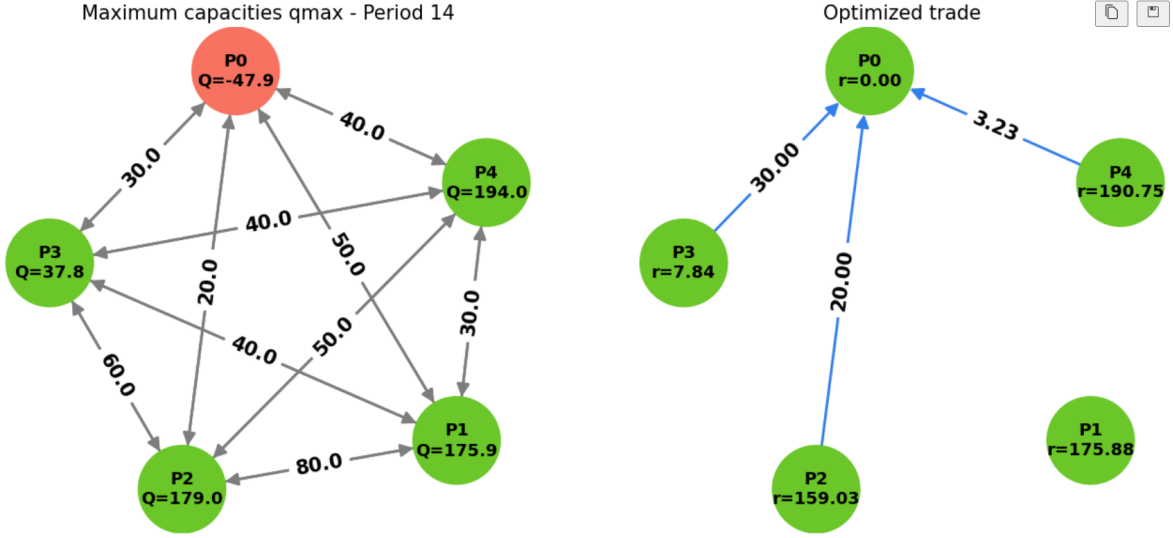


Figure 7: Example 2 with 5 countries: Trade graphs at one hour t_2

Figure 5 shows that the exchanges did reduce the deficits for each country, often to zero in this case.

Figures 6 and 7 illustrate the solution provided by our program for a given time t . The graph on the left shows the amount of electricity available (positive or negative) at time t for each country ($P0$ to $P4$) and the maximum amount of electricity that can be exchanged between each country (indicated by gray arrows). The graph on the right shows the amount of electricity available for each country after the exchanges and the amount exchanged (blue arrows).

2.3. Storage problem. This problem primarily aims to minimize deficits while maximizing the amount of electricity stored. We seek to optimize the different storage methods, hourly, over a given period, for a single country. The cost function is the sum of these two optimizations with a penalty σ . When $\sigma \rightarrow 0$, only the level of electricity stored at the end of the period is important, whereas when $\sigma \rightarrow \infty$, only minimizing shortages is important. You can see the optimization problem just below.

$$\min \quad -\text{level}_{\text{PTG}}^{T-1} + \sigma \cdot \sum_{t=1}^T \text{deficit}^t,$$

under the constraints :

$$\begin{aligned}
\text{level}_{str}^0 &= \text{capacity}_{str} \times \text{fillrate}_{str}^0, \\
\text{powerin}_{str}^t &\leq \text{powerin}_{str}^{\max}, \\
\text{powerout}_{str}^t &\leq \text{powerout}_{str}^{\max}, \\
\text{level}_{str}^t &\leq \text{capacity}_{str}, \\
\text{prod}^t - \text{demand}^t + \sum_{str} (\text{powerout}_{str}^t \times \text{efficiency}_{str} - \text{powerin}_{str}^t) \\
&= \text{surplus}^t - \text{deficit}^t, \\
\text{level}_{str}^t &= \text{level}_{str}^{t-1} \\
&\quad + \text{powerin}_{str}^t \times \text{efficiency}_{str} \\
&\quad - \text{powerout}_{str}^t.
\end{aligned}$$

We obtain the following results, presented in Figures 8 and 9.

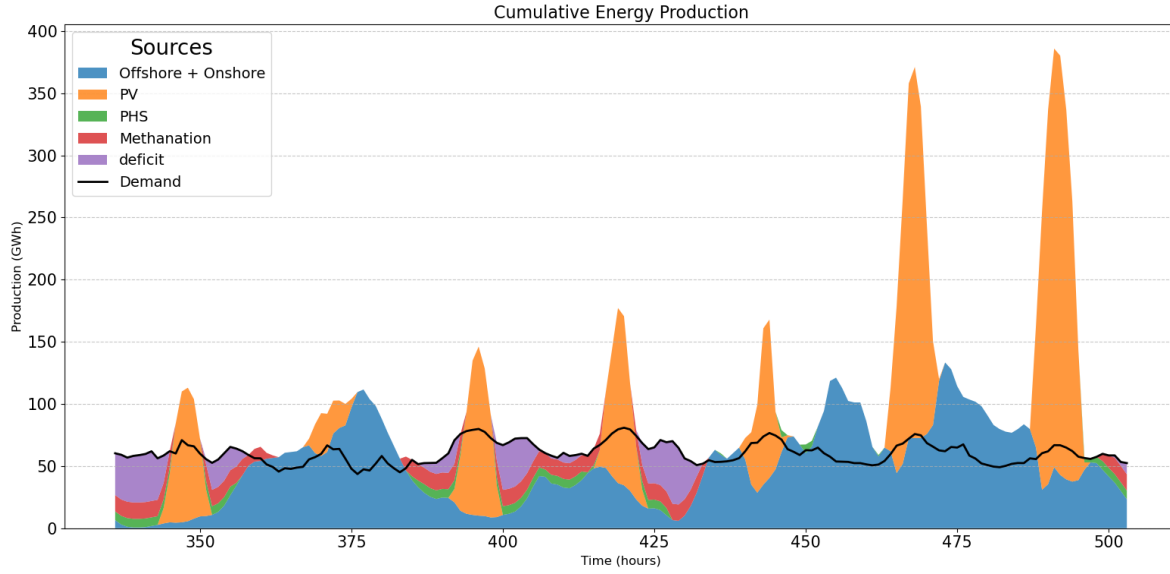


Figure 8: Electricity production over a typical week

Figure 8 shows the electricity production in GWh of the production and storage methods studied in this model. The black curve represents France's electricity demand, and the colored curve represents the production of the different generation methods. The purple curve shows the deficits resulting from the predefined electricity mix. We can see the day/night cycle with the peaks of solar production in orange.

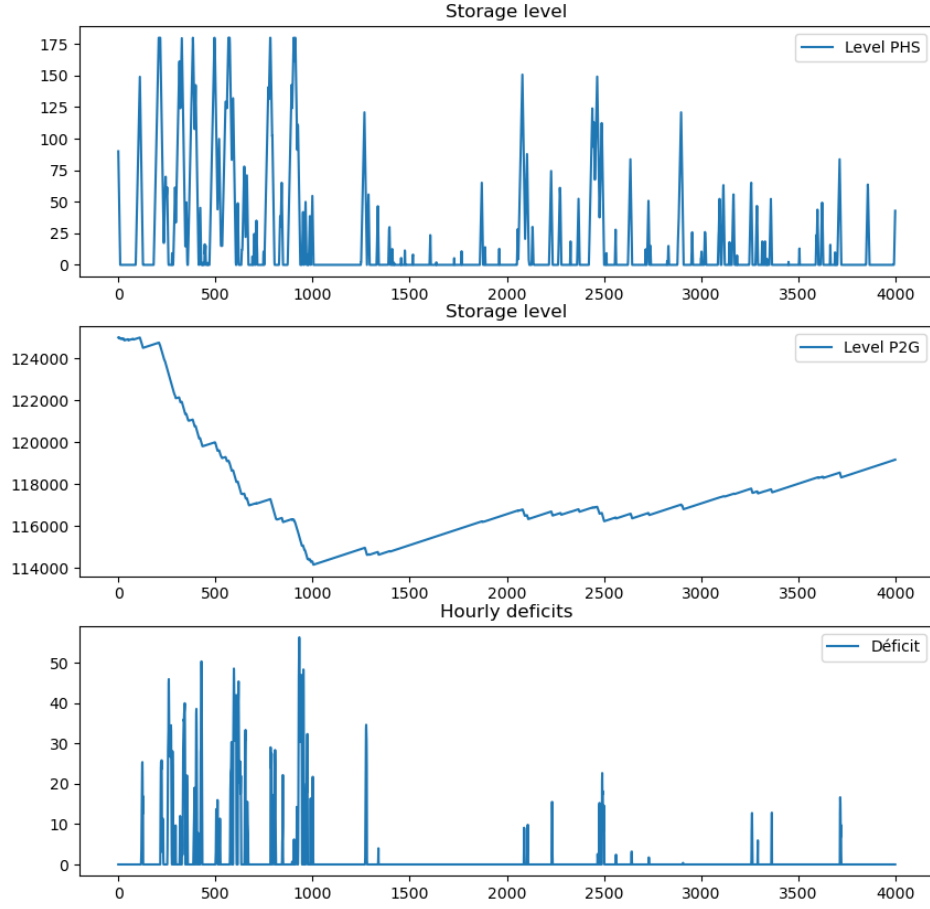


Figure 9: Deficit and amount of electricity stored per technology over a year

Figure 9 shows curves illustrating the variables optimized by the model. We observe that the PHS storage level changes frequently with very rapid spikes; this is due to the fact that energy is stored and released quickly in reality. Conversely, the gas storage level evolves slowly, either positively or negatively, illustrating the slow storage and release of gas.

2.4. Simulation. Below you can see graphs obtained from the simulations of the heuristic approach. These results will be compared to the reference in the next section.

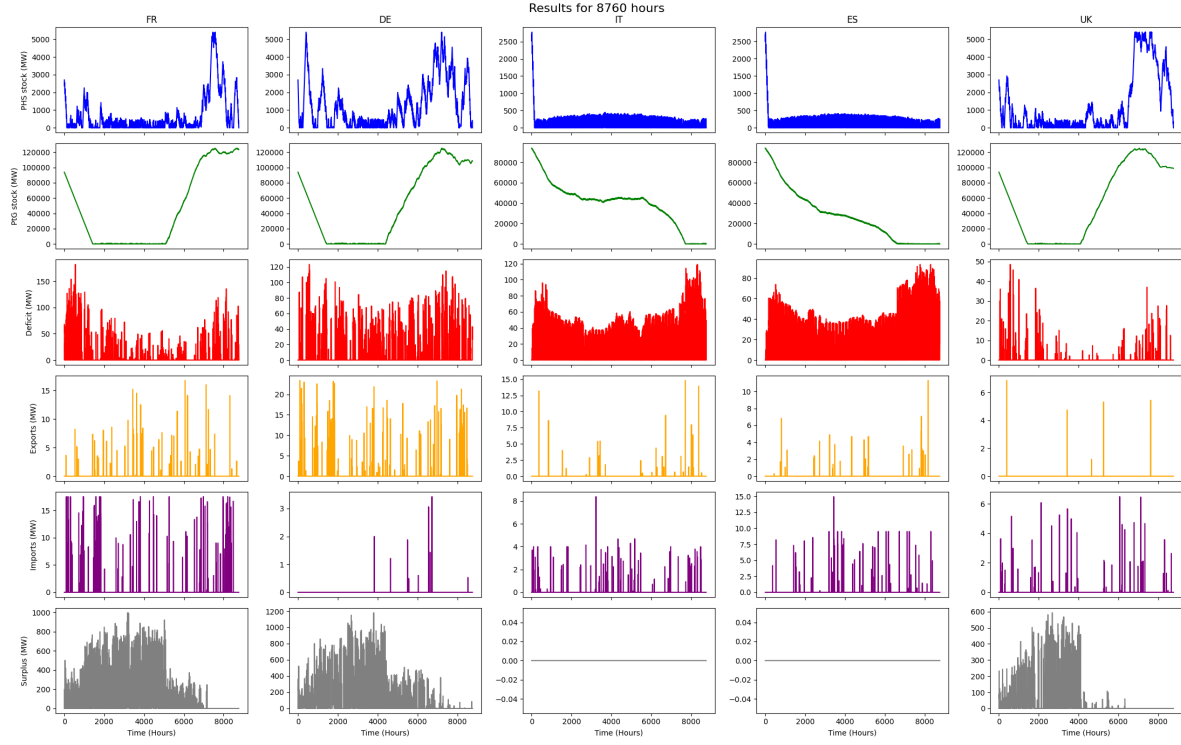


Figure 10: Optimization results - PHS level, PTG level, deficit, imports, exports and surplus for each country over one year

In Figure 10, the observations are broadly the same as in Figure 2. However, we can see that there are way less imports and exports. Moreover, it is noticeable that the quantities of deficit are significantly larger and that Spain and Italy are unable to replenish their PTG stocks without increasing the deficit considerably.

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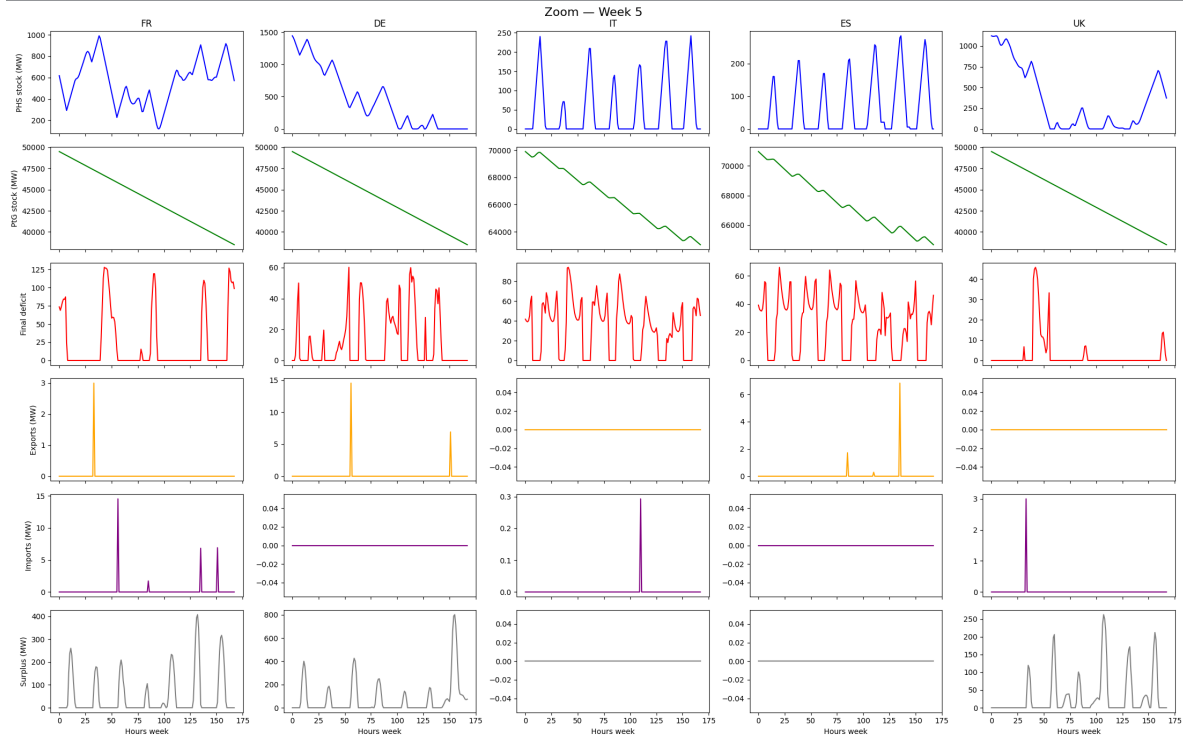


Figure 11: Optimization results - PHS level, PTG level, deficit, imports, exports and surplus for each country over one week

In Figure 11, we can see a zoomed-in view of the first week of the year. Not many exchanges took place this week. This is probably due to the fact that, as exchange optimization is initially done on an hourly basis, all countries need electricity at the same time, so no exchanges are made.

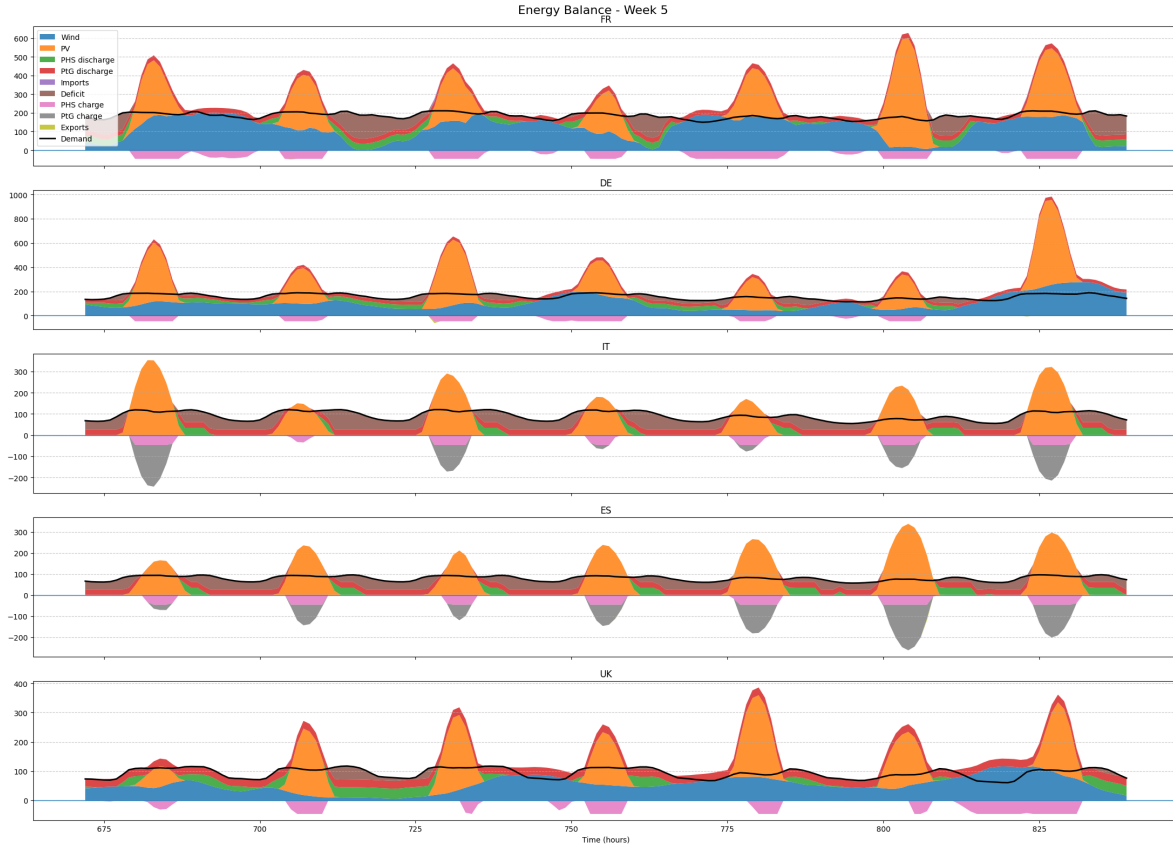


Figure 12: Cumulative graph of electricity quantities and demand curve - week 1

Finally, Figure 12 confirms the previous results: there are not many imports/exports and that results in large deficit.

3. Results and discussion. The first thing we can say is that we have succeeded in considerably reducing computation time: from an average of 60 seconds to 15 seconds, which is a significant runtime reduction. Even if the results obtained by the heuristic model are not exactly the same as the ones from the complete model, they met our expectations as you can see in Figure 13.

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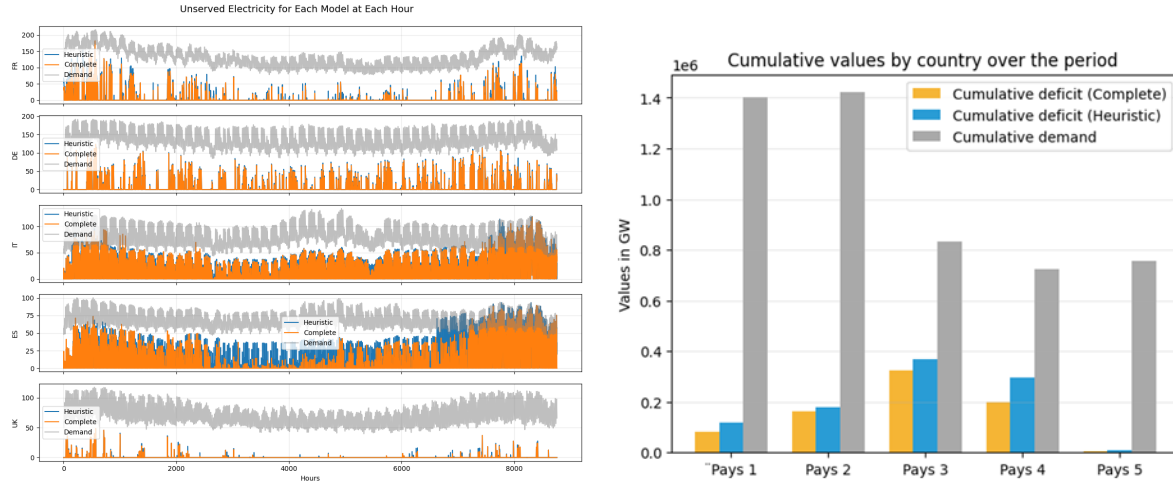


Figure 13: Comparison of the deficit in the two approaches

In Figure 13, we can see that the deficit level is higher in the heuristic model (blue), particularly in Spain. As mentioned previously, these results are due to the order in which the exchanges were processed. The exchanges are structured so that each country has its share of electricity at each hour within a tolerance interval representing the amount that can be loaded or unloaded from storage technologies. This method completely disregards whether or not a country possesses stored electricity. This choice was made because, when optimizing the exchanges, the tolerance interval does not need to be constantly updated. Furthermore, at this stage, the optimization of storage strategies has not yet been performed.

Conclusion. In this study, we developed a new approach to the reference solution established by the complete algorithm. This new algorithm makes storage optimization independent of electricity exchanges. As a result, we have managed to reduce the program execution time by a factor of four. In our experiments, the average difference in the share of unmet demand was 4.8%, which is reasonable.

The main limitation of this algorithm is that it cannot take into account the stock level (of PHS or PTG) when implementing the electricity exchange strategy. Therefore, it does not lead to imports or exports due to the anticipation of future needs.

Moreover, we used synthetic data to first test whether our algorithm was working properly. A next step could be to retrieve real data (for demand, for example) but also to test the reliability of the algorithm in extreme weather events.

Finally, the objective would be to extend this method to a more complex problem because it is where the number of variables leads to a need for reduced program execution time.

Acknowledgments. We used the AI Claude to help us formulate the abstract and introduction, and to code part of our linear programming algorithm.

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